Inorganic and Materials Cumulative Exam
October 20, 2017

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

1. (a) (14pts) Using a molecular orbital diagram, describe the electronic structure of LiH (Li 1s orbital potential = -56eV, Li 2s orbital potential = -5.4eV, H 1s orbital potential = -13.6eV).

(b) (6pts) Estimate the polar nature of any bonding or antibonding molecular orbitals formed.

2. (a) (5pts) How many vibrational modes should one expect from SO_2?

(b) (15pts) Sketch each vibrational mode of SO_2, indicating motions with arrows and/or description as needed.

3. The existence of high-spin octahedral complexes is an apparent violation of the Aufbau principle.

(a) (4pts) What is the Aufbau principle?

(b) (10pts) Draw the frontier molecular orbital diagram of a d^6 octahedral complex in a high-spin configuration.

(c) (6pts) How do electron-electron interactions explain why high-spin configurations are stable?

4. (a) (15pts) Draw the structure of a ReH_9^{2-} ion.

(b) (5pts) What property of H^+ enables it to form a stable complex with Re with a coordination number of 9?

5. (a) (4pts) What is the point group of CO_2?

(b) (8pts) Using the included character table (descending in symmetry), how many IR-active modes does CO_2 have?

(b) (8pts) Using the included character table (descending in symmetry), how many Raman-active modes does CO_2 have?

6. (a) (10pts) Draw a hypothetical discharge curve for a battery (% charge vs. cell voltage) at very low current.

(b) (5pts) On the same graph, draw a hypothetical discharge cure for a battery at high current.

(c) (5pts) In terms of the electrochemistry of the cell, explain the difference in the two curves.

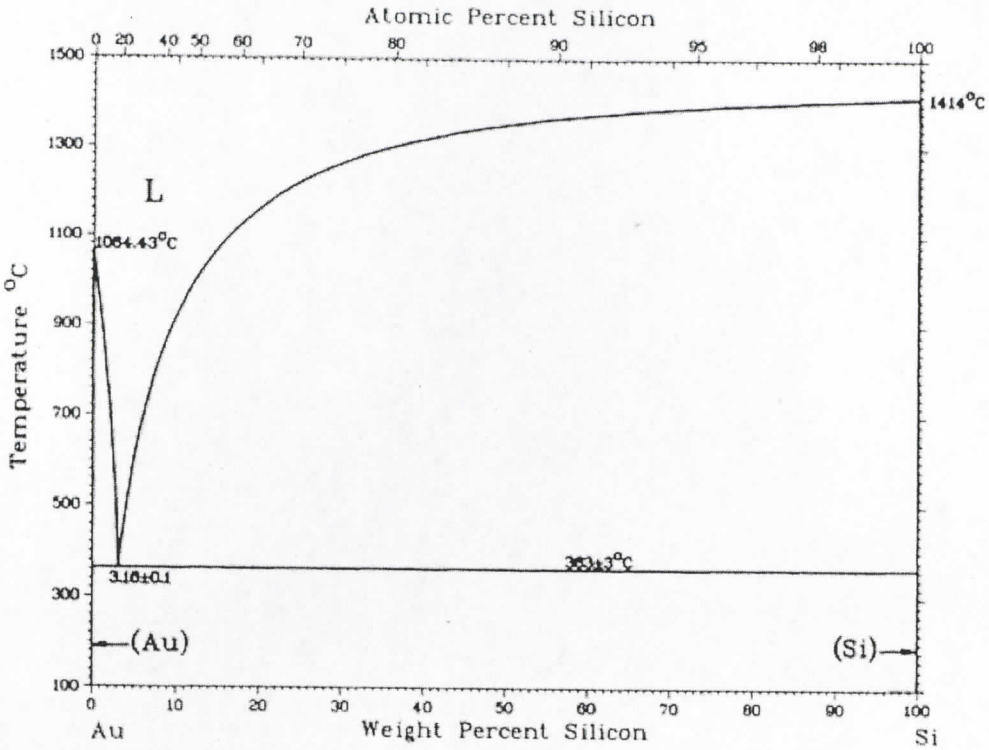
7. (a) (12pts) Explain a van der Waals attraction in terms of electrostatics.

(b) (8pts) Is this different than ionic bonding? Why or why not?

8. (a) (7pts) Define the term "eutectic system" for a binary alloy.

(b) (7pts) Identify the eutectic composition and eutectic temperature for the Au-Si system from the included phase diagram.

(c) (6pts) What happens to an alloy of Au-Si at the eutectic composition at temperatures above the eutectic temperature?



Au-Si phase diagram

9. Use description and band diagrams to describe the construction and operation of a bipolar transistor.

10. (a) (15 pts) Define the isotropic, nematic, and smectic phases of a liquid crystal.

(b) (5pts) Rank the phases in order of 'most ordered' to 'least ordered'?

$$\# reps = \frac{1}{h}\sum(g * \chi_r * \chi_i)$$

h = total operations in a group

g = number of operations in a class

χ_r = the character of the reducible representation

χ_i= the character of the irreducible representation

The transformation matrix for a C_n rotation around the z axis is

$$\begin{bmatrix} cos\,\theta & -sin\theta & 0 \\ sin\theta & cos\,\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$cos\frac{2\pi}{1} = 1, sin\frac{2\pi}{1} = 0$

$cos\frac{2\pi}{2} = -1, sin\frac{2\pi}{2} = 0$

$cos\frac{2\pi}{3} = -0.5, sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

$cos\frac{2\pi}{4} = 0, sin\frac{2\pi}{4} = 1$

$cos\frac{2\pi}{5} = \frac{1}{4}(-1+\sqrt{5}), sin\frac{2\pi}{5} = \frac{1}{4}\sqrt{10+2\sqrt{5}}$

$cos\frac{2\pi}{6} = 0.5, sin\frac{2\pi}{6} = \frac{\sqrt{3}}{2}$

Character Tables

C_2	E	C_2		
A	1	1	z, R_z	x^2, y^2, z^2, xy
B	1	-1	x, y, R_x, R_y	yz, xz

C_3	E	C_3	$(C_3)^2$		
A	1	1	1	z, R_z	x^2+y^2, z^2
E	1 1	e e^*	e^* e	x+iy; R_x+iR_y x-iy; R_x-iR_y	(x^2-y^2, xy) (yz, xz)

e = exp(2πi/3)

C_4	E	C_4	C_2	$(C_4)^3$		
A	1	1	1	1	z, Rz	x^2+y^2, z^2
B	1	-1	1	-1		x^2-y^2, xy
E	1 1	i -i	-1 -1	-i i	x+iy; R_x+iR_y x-iy; R_x-iR_y	(yz, xz)

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	-1	-1	R_z	xy
B_1	1	-1	1	-1	x, R_y	xz
B_2	1	-1	-1	1	y, R_x	yz

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	-1	R_z	
E	2	-1	0	(x, y) (R_x, R_y)	(x^2-y^2, xy) (xz, yz)

C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		x^2-y^2
B_2	1	-1	1	-1	1		xy
E	2	0	-2	0	0	(x, y) (R_x, R_y)	(xz, yz)

C_{6v}	E	$2C_6$	$2C_3$	C_2	$3\sigma_v$	$3\sigma_d$		
A_1	1	1	1	1	1	1	z	x2+y2, z2
A_2	1	1	1	1	-1	-1	Rz	
B_1	1	-1	1	-1	1	-1		
B_2	1	-1	1	-1	-1	1		
E_1	2	1	-1	-2	0	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	=1	=1	2	0	0		(x^2-y^2, xy)

C_{5v}	E	$2C_5$	$2(C_5)^2$	$5\sigma_v$		
A_1	1	1	1	1	z	x2+y2, z2
A_2	1	1	1	-1	Rz	
E_1	2	2cos(2π/5)	2cos(4π/5)	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	2cos(4π/5)	2cos(2π/5)	0		(x2-y2, xy)

D_{2h}	E	C_2 (z)	C_2 (y)	C_2 (x)	i	σ (xy)	σ (xz)	σ (yz)		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	-1	-1	1	1	-1	-1	R_z	xy
B_{2g}	1	-1	1	-1	1	-1	1	-1	R_y	xz
B_{3g}	1	-1	-1	1	1	-1	-1	1	R_x	yz
A_u	1	1	1	1	-1	-1	-1	-1		
B_{1u}	1	1	-1	-1	-1	-1	1	1	z	
B_{2u}	1	-1	1	-1	-1	1	-1	1	y	
B_{3u}	1	-1	-1	1	-1	1	1	-1	x	

D_{3h}	E	$2C_3$	$3C'_2$	σ_h	$2S_3$	$3\sigma_v$		
A'_1	1	1	1	1	1	1		x^2+y^2, z^2
A'_2	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A''_1	1	1	1	-1	-1	-1		
A''_2	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

D_{4h}	E	$2C_4$	C_2	$2C'_2$	$2C''_2$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	-1	-1	1	1	1	-1	-1	Rz	
B_{1g}	1	-1	1	1	-1	1	-1	1	1	-1		x^2-y^2
B_{2g}	1	-1	1	-1	1	1	-1	1	-1	1		xy
E_g	2	0	-2	0	0	2	0	-2	0	0	(Rx, Ry)	(xz, yz)
A_{1u}	1	1	1	1	1	-1	-1	-1	-1	-1		
A_{2u}	1	1	1	-1	-1	-1	-1	-1	1	1	z	
B_{1u}	1	-1	1	1	-1	-1	1	-1	-1	1		
B_{2u}	1	-1	1	-1	1	-1	1	-1	1	-1		
E_u	2	0	-2	0	0	-2	0	2	0	0	(x, y)	

T_d	**E**	**$8C_3$**	**$3C_2$**	**$6S_4$**	**$6\sigma_d$**		
A_1	1	1	1	1	1		$x^2+y^2+z^2$
A_2	1	1	1	-1	-1		
E	2	-1	2	0	0		$(2z^2-x^2-y^2, x^2-y^2)$
T_1	3	0	-1	1	-1	(R_x, R_y, R_z)	
T_2	3	0	-1	-1	1	(x, y, z)	(xy, xz, yz)